Towards a Next-Generation Agricultural Expert System: A Multimodal Conversational Approach

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Introduction: Next-Generation Agricultural Expert System

- **The Shift:** Moving from static, rule-based apps to AI-driven conversational agents.
- **The Challenge:** Farming requires synthesizing heterogeneous data (visual anomalies, complex soil chemistry, local policies).
- **Current Limitations:** Generic LLMs hallucinate agronomic advice, lack domain specificity, and fail on low-quality field media.
- **Proposed Solution:** A multimodal framework combining computer vision, dynamic knowledge retrieval, and strict domain constraints.



Related Work – Vision and Multimodality

- **Multimodal LLMs (e.g., AgriM-LLM):** Attempt to fuse visual encoders with language models for disease management.
  - *Limitation:* Accuracy degrades heavily on the blurry, compressed media typical of real-world farming.
- **Diagnostic CNNs:** Proven highly accurate in identifying nutrient deficiencies via static plant imagery.
  - *Limitation:* Over-reliant on high-fidelity, static images rather than dynamic, noisy field video.



Related Work – Architecture and NLP

- **Retrieval-Augmented Generation (RAG):** The current standard for grounding LLMs in localized data to prevent hallucinations.
  - *Limitation:* Current RAG implementations lack temporal memory; they treat queries in isolation instead of tracking a full crop cycle.
- **Lexical Processing:** Standard NLP pipelines often fail to capture localized agricultural vernacular and context.
- **The Regulatory Void:** Existing chatbots offer chemical/agronomic advice while operating in a legal vacuum, completely bypassing agricultural laws and compliance frameworks.

Identified Research Gaps

1. **Low-Fidelity Media Handling:** Current models cannot accurately classify anomalies from the low-resolution, noisy images or videos captured by standard smartphones in the field.

2. **Contextual Memory Loss:** Existing RAG architectures fail to maintain a continuous, evolving state of a farm over an entire harvesting season.

3. **The Compliance Disconnect:** There is an absence of expert systems that cross-reference deep biochemical recommendations with regional agricultural regulations and legal constraints.



Research Objectives

1. **Enhance Visual Input:** Integrate Super-Resolution (Image and Video) preprocessing to restore low-fidelity field media prior to diagnostic classification.

2. **Develop Robust Architecture:** Construct an advanced RAG framework equipped with state-tracking to maintain full-season agronomic context.

3. **Embed Multidisciplinary Intelligence:** Bridge agronomic advice with legal compliance, ensuring all soil and chemical recommendations adhere strictly to local policies.



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Thank You!

Any Questions?

